6. Develop an Android application demonstrating communication between Activity and Fragment.

Project Name: ActivityFragmentCommunication

Creating Fragment
java package → New→ Fragment → Fragment (Blank)
Fragment Name: DemoFragment
Layout Name: fragment_demo

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<FrameLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:id="@+id/fragment_container"

   android:layout_width="match_parent"

   android:layout_height="match_parent"/>
```

Open From res/layout:
Filename: fragment_demo.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:gravity="center"

   android:orientation="vertical"

   android:padding="20dp">



   <TextView
```

```
       android:id="@+id/txtMessage"

       android:layout_width="wrap_content" android:layout_height="wrap_content"
android:textSize="22sp"

       android:text="Message from Activity"/>



   <Button

       android:id="@+id/btnSend"

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Send Message to Activity"

       android:layout_marginTop="20dp"/>

</LinearLayout>
```

Modify DemoFragment.java.

Filename: DemoFragment.java

```
package com.example.activityfragmentcommunication;


import android.content.Context;

import android.os.Bundle;

import android.view.LayoutInflater;

import android.view.View;

import android.view.ViewGroup; import android.widget.Button;

import android.widget.TextView;

import androidx.annotation.NonNull;

import androidx.annotation.Nullable;
```

```
import androidx.fragment.app.Fragment;

public class DemoFragment extends Fragment

{

   TextView txtMessage;

   Button btnSend;

   String activityMessage;



   FragmentCommunication communication;

public interface FragmentCommunication {

   void onMessageFromFragment(String message);

}



// Receive Activity object

@Override

public void onAttach(@NonNull Context context) {

   super.onAttach(context);

   communication = (FragmentCommunication) context;

}







// Method to receive data from Activity

public static DemoFragment newInstance(String message) {

   DemoFragment fragment = new DemoFragment();

   Bundle bundle = new Bundle();

   bundle.putString("msg", message);
```

```
    fragment.setArguments(bundle);

    return fragment;

}




@Override

public View onCreateView(LayoutInflater inflater,ViewGroup container,Bundle
savedInstanceState) {

    View view = inflater.inflate(R.layout.fragment_demo,container, false);

    txtMessage = view.findViewById(R.id.txtMessage);

    btnSend = view.findViewById(R.id.btnSend);

// Receiving data from Activity

    if(getArguments() != null){

        activityMessage = getArguments().getString("msg");

        txtMessage.setText(activityMessage);

    }



// Sending data to Activity

    btnSend.setOnClickListener(v -> {

        communication.onMessageFromFragment(

                "Hello Activity! Message from Fragment");

    }

    );



        return view;

}
```

```
}
```

Filename: MainActivity.java

```
package com.example.activityfragmentcommunication;


import android.os.Bundle;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity

       implements DemoFragment.FragmentCommunication {


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);


       // Sending data from Activity to Fragment

       DemoFragment fragment = DemoFragment.newInstance(

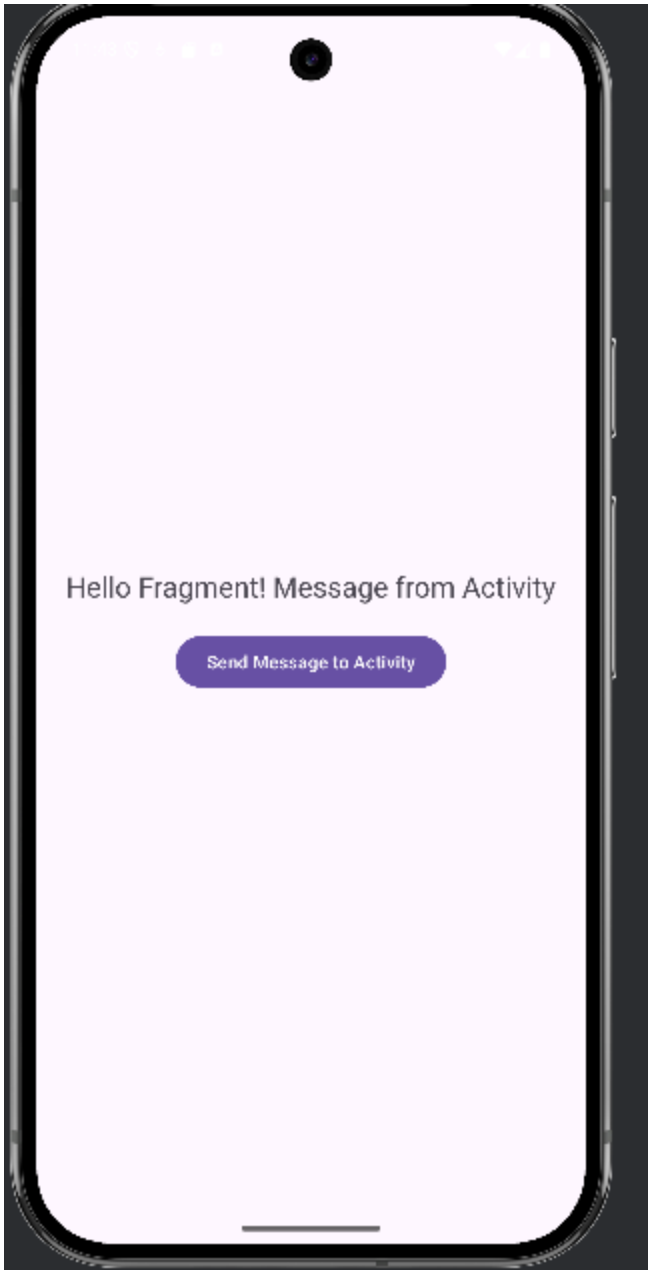
               "Hello Fragment! Message from Activity"
```

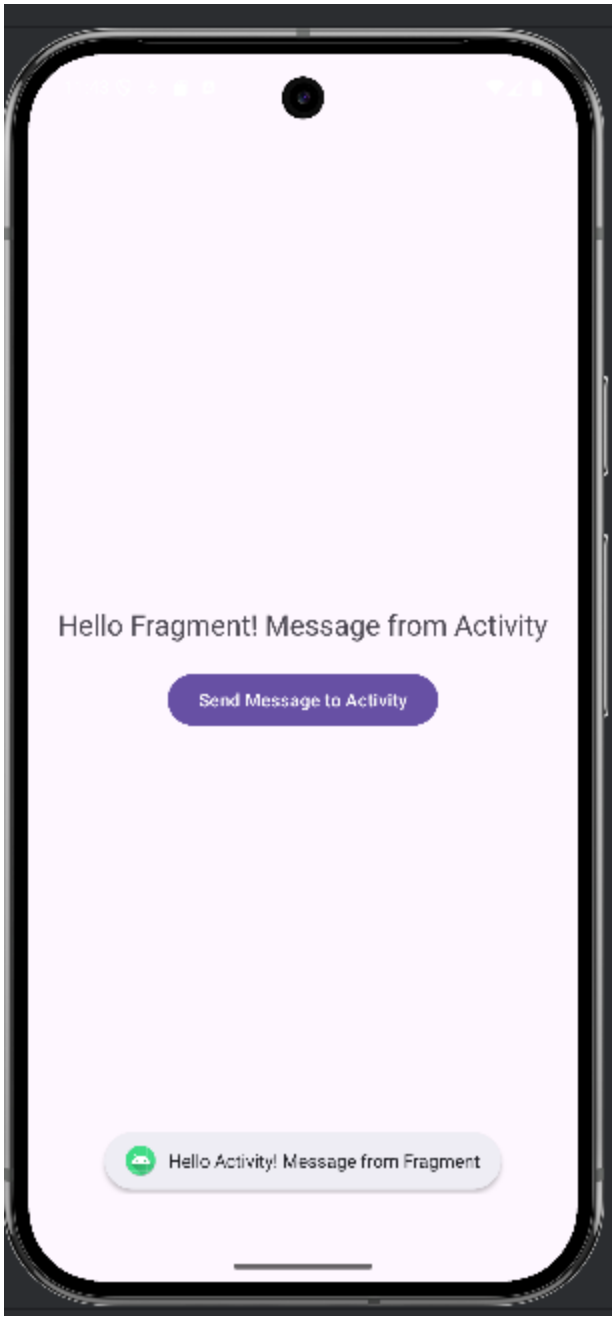
```
        );


        getSupportFragmentManager()
                .beginTransaction()
                .replace(R.id.fragment_container, fragment)
                .commit();
    }


    // Receiving data from Fragment
    @Override
    public void onMessageFromFragment(String message) {
        Toast.makeText(this, message, Toast.LENGTH_LONG).show();
    }
}
```

OUTPUT:


Hello Fragment! Message from Activity
Send Message to Activity


Hello Fragment! Message from Activity
Send Message to Activity
Hello Activity! Message from Fragment